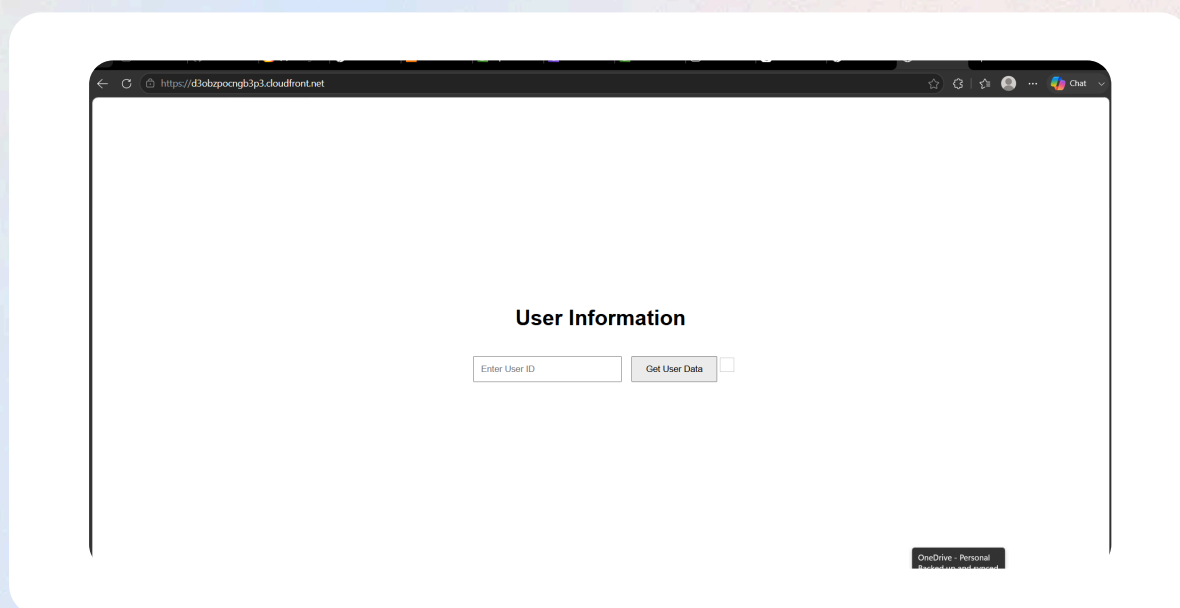


# WEBSITE DELIVERY WITH CLOUDFRONT

By Mohd Mudassir Arfath | July 2026

## Website Delivery with CloudFront



# Introducing Today's Project!

In this project, I will demonstrate... I'm doing this project to learn...

## Tools and concepts

In this project, I learned how to use Amazon S3 to store website files and Amazon CloudFront to deliver website content quickly through a global network of edge locations. I also learned about CloudFront distributions, origins, Origin Access Control (OAC), S3 bucket policies, permissions, and how to troubleshoot Access Denied errors. Overall, I learned how S3 and CloudFront work together to securely deliver a website.

## Project reflection

It took me approximately 2 to 3 hours to complete this project, including setting up the S3 bucket, uploading the website files, configuring CloudFront, updating permissions, and troubleshooting the Access Denied error.

Thank you! I enjoyed completing this project and learned a lot about Amazon S3, CloudFront, Origin Access Control, and secure website delivery.

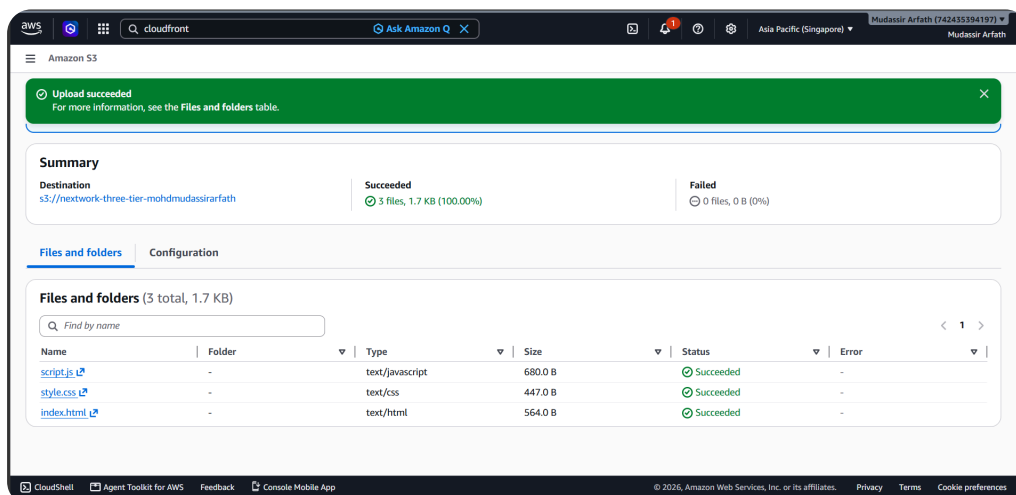
## Set Up S3 and Website Files

We are using an S3 bucket to store and host the static website files, such as HTML, CSS, JavaScript, and images. Amazon CloudFront uses the S3 bucket as the origin to deliver the website content quickly and securely to users.

The three files I downloaded were:

1. `index.html`
2. `style.css`
3. `script.js`

I validated that I validated my website files by checking that all required files were successfully uploaded to the S3 bucket. I also verified that the files were accessible and that the website content was displayed correctly.  
my website files work by...

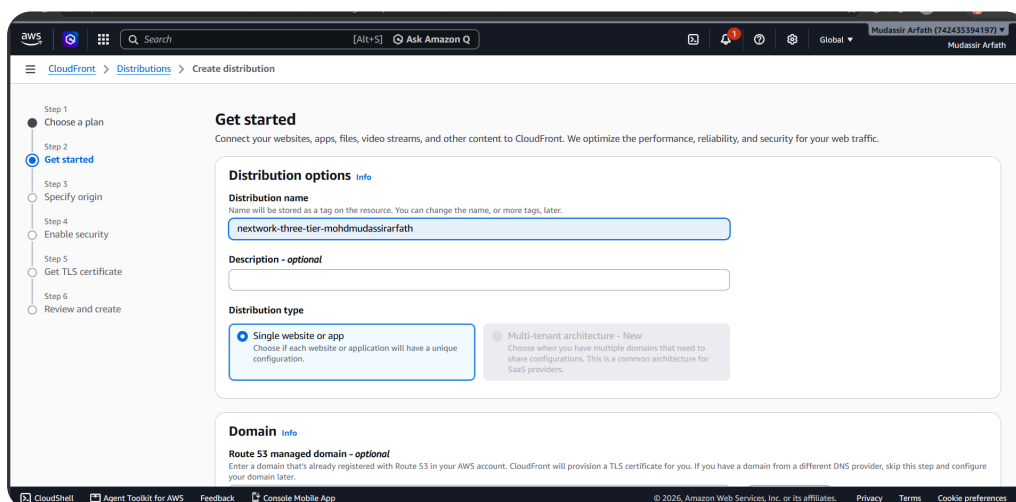


# Exploring Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) service that delivers website content quickly and securely to users through a global network of edge locations. It helps reduce latency and improves website performance.

A CloudFront distribution is a configuration that tells Amazon CloudFront how to deliver content to users. It connects CloudFront to an origin, such as an S3 bucket, and distributes the content through edge locations around the world.

I set up the CloudFront distribution by selecting the S3 bucket as the origin and configuring the required CloudFront settings. I then created the distribution and waited for it to deploy so that the website could be delivered through the CloudFront distribution.

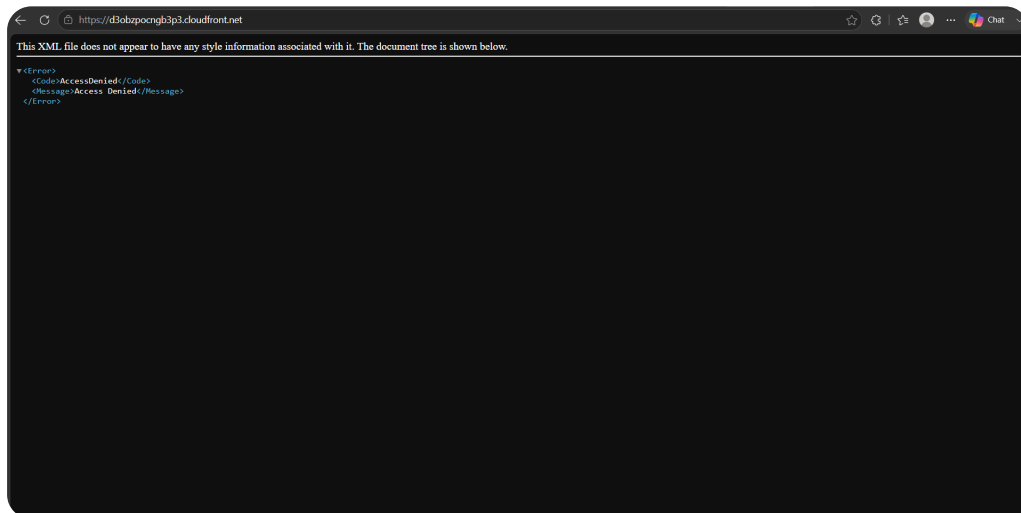


## Handling Access Issues

I ran into the Access Denied error because the CloudFront distribution was not configured with the correct access permissions to access the S3 bucket. The origin settings were not properly configured, so CloudFront could not retrieve the website files from the S3 bucket.

My CloudFront distribution used Origin Access Control (OAC) to securely access the S3 bucket. The S3 bucket was configured to allow access only through the CloudFront distribution.

Origin Access Control (OAC) is a CloudFront feature that allows CloudFront to securely access content stored in an Amazon S3 bucket. It helps prevent direct public access to the S3 bucket and ensures that users access the website through CloudFront.



## Updating S3 Permissions

A bucket policy update is still needed to give the CloudFront distribution permission to access and retrieve the website files from the S3 bucket. Without the correct bucket policy, CloudFront cannot access the S3 objects, which can result in an Access Denied error.

This policy allows the CloudFront distribution to access and retrieve the website files stored in the S3 bucket. It gives CloudFront permission to read the objects from the bucket while keeping the S3 bucket private and secure.

